MAPÚA UNIVERSITY

Laboratory Report 1

Software Relevant Tools, Standards, and Code Versioning using Github

Name:	Mones, Quert Benedict V.	**Date:**	August 22, 2026
Program:	CPE	**Section:**	B4

A. Learning Outcomes:
At the end of this laboratory, students should be able to:

1. Identify software development tools and standards used in software projects.
2. Create and manage a GitHub repository.
3. Apply Git version control commands.
4. Collaborate using GitHub features.
5. Demonstrate proper documentation and coding standards.

B. Laboratory Scenario:

You have been hired as a junior software developer in a software company. Your team uses GitHub for source code management and follows coding standards to maintain software quality. Your task is to create a project repository, manage versions of source code, and document your work according to industry standards.

C. Practical Tasks:

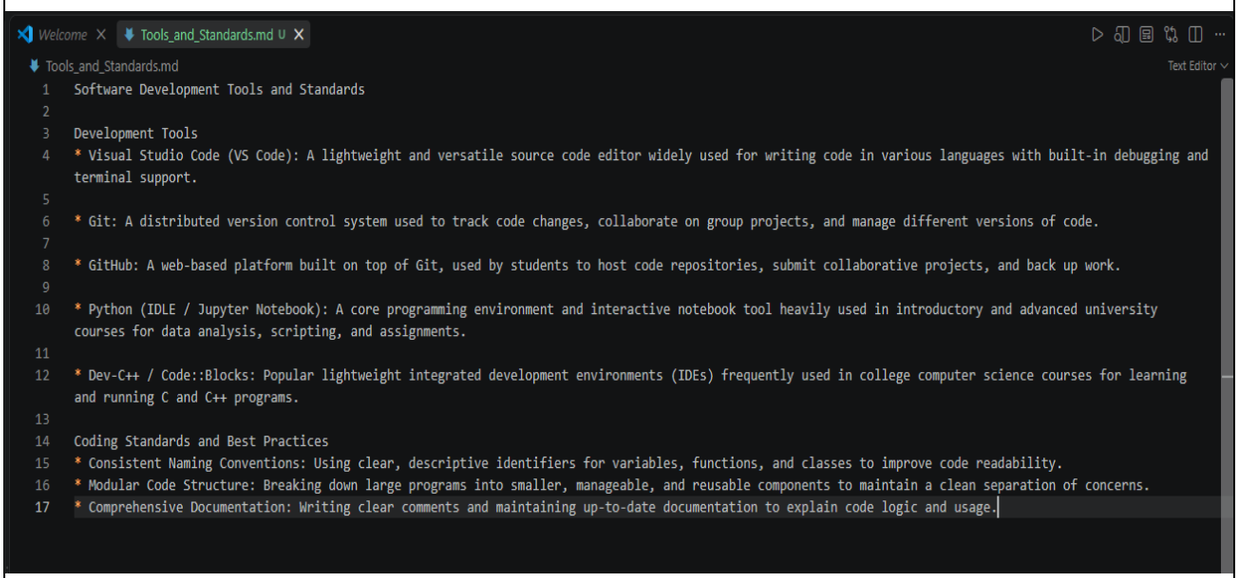
1. Software Development Tools and Standards

Create a document named Tools_and_Standards.md containing:

a. List at least 5 development tools.
b. Describe the purpose of each tool.
c. Identify at least three coding standards or best practices used in software development.

(Please include the following: **1.** Screenshots of the output)

Screenshot of Output
(Insert images here from the output)



2. GitHub Repository Creation
   a. Create a GitHub account (if none exists).
   b. Create a public repository named: Software-Versioning-Lab
   c. Add the following:
      i. README.md
      ii. LICENSE
      iii. .gitignore
   d. Upload this laboratory file in this repository once accomplished.
   e. Submit GitHub repository link
   f. Screenshot of repository homepage

Laboratory Report 1

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(Please include the following: **1.** Screenshots of the output)

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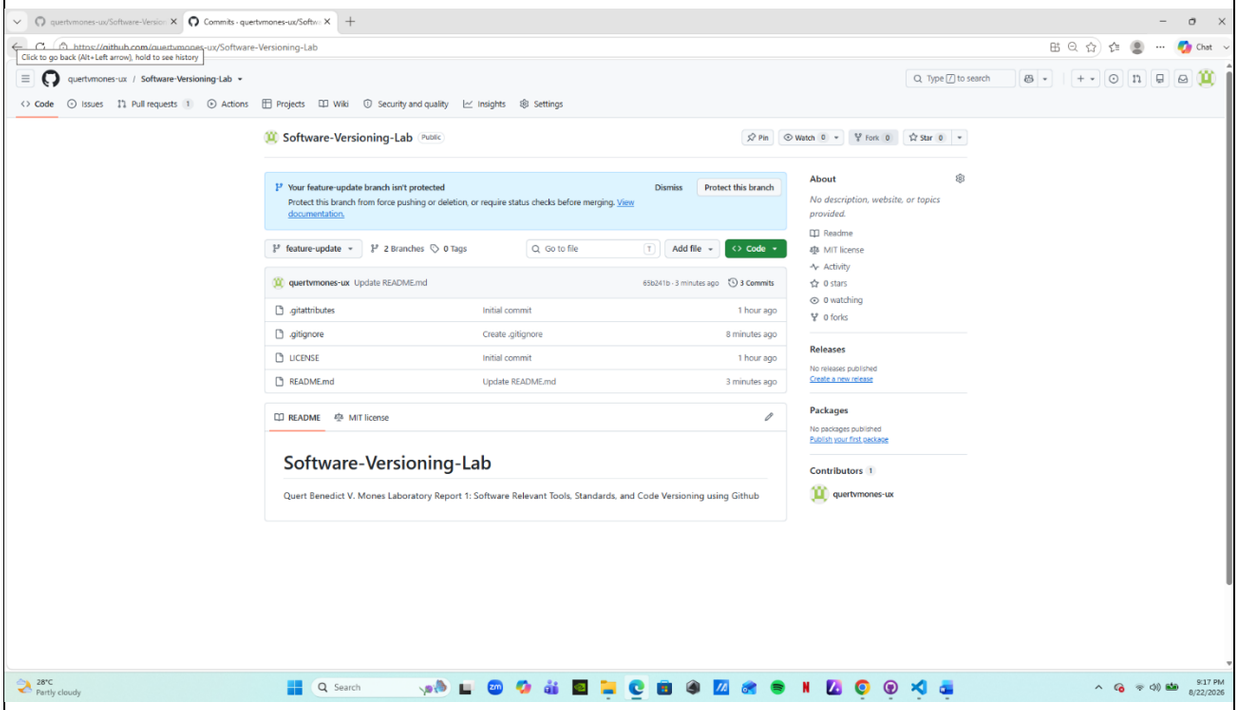
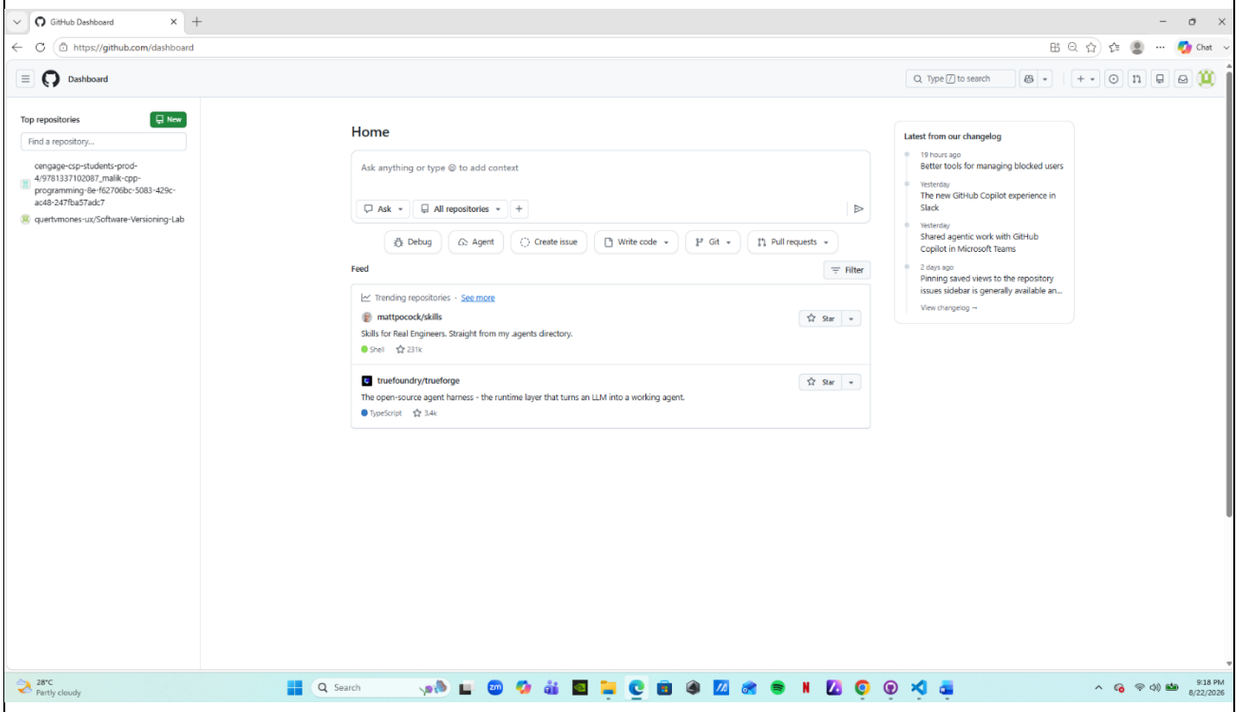
(Insert images here from the output)





quertvmones-ux/Software-Versioning-Lab

3. Git Version Control
    a. Perform the following Git commands and document them in a file named: (Git_Report.md)
        i. Clone Repository
            a) git clone <repository-url>
        ii. Creating New Branch
            a) git checkout -b feature-update

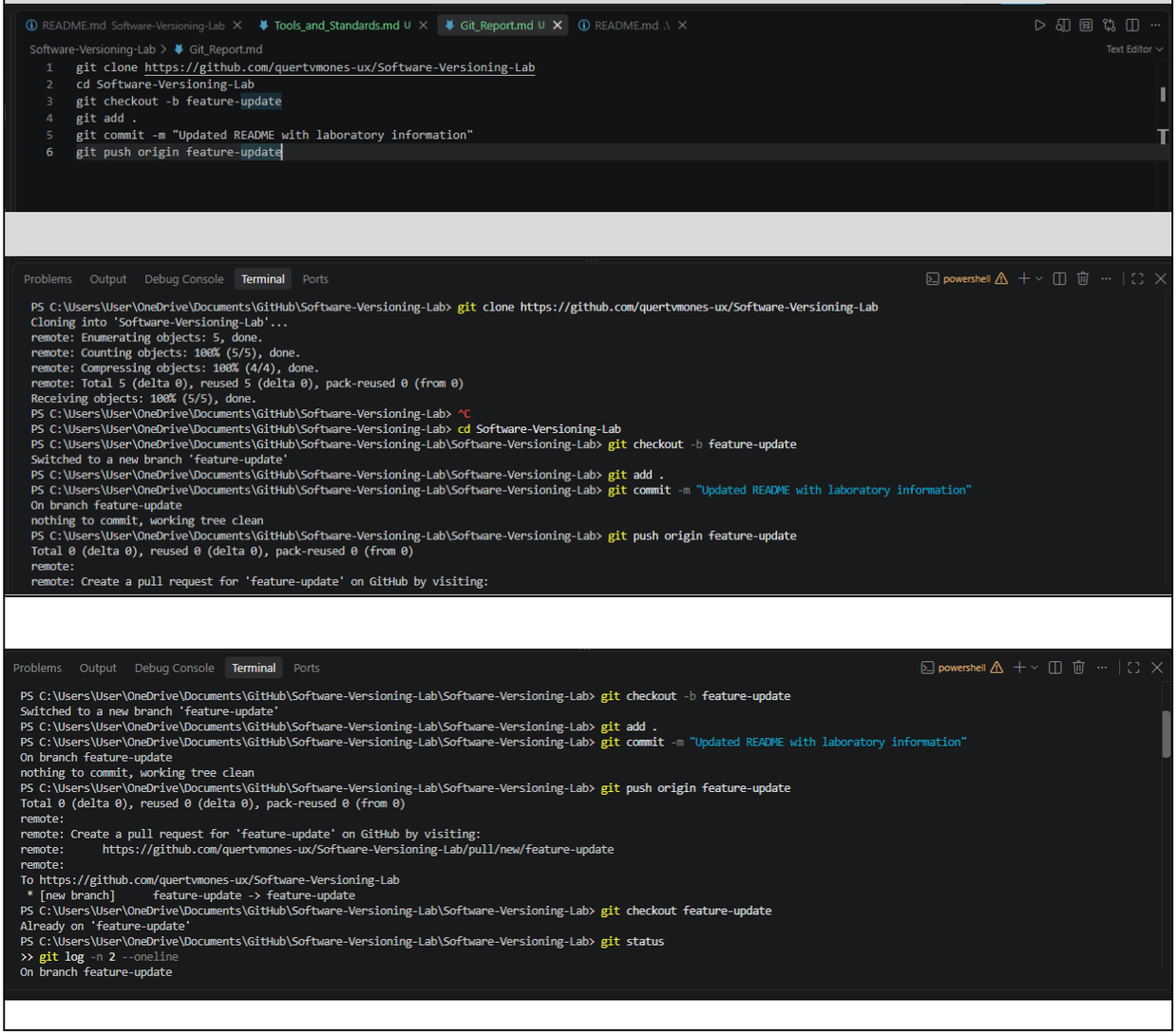
Software Relevant Tools, Standards, and Code Versioning using Github

iii. Modify README.md
   a) Add your name and laboratory information
iv. Commit Changes
   a) git add .
   b) git commit -m "Updated README with laboratory information"
v. Push Changes
   a) git push origin feature-update
vi. Create Pull Request
   a) feature-update → main

b. Screenshot the following:
   i. Commands executed
   ii. Pull Request
   iii. Commit history

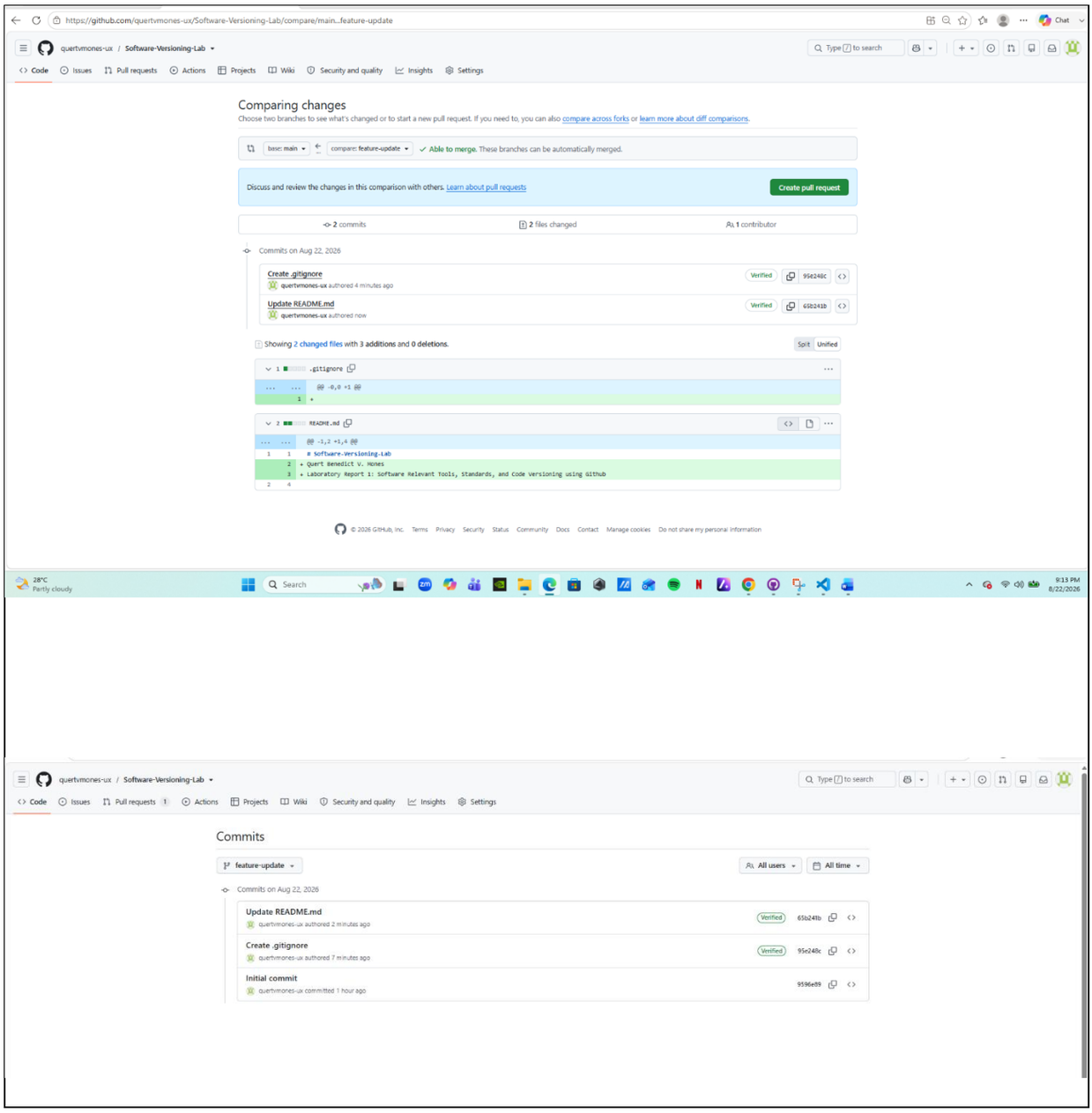
(Please include the following: **1.** Screenshots of the output)

Screenshot of Output



Laboratory Report 1

Software Relevant Tools, Standards, and Code Versioning using Github



Laboratory Report 1

Software Relevant Tools, Standards, and Code Versioning using Github

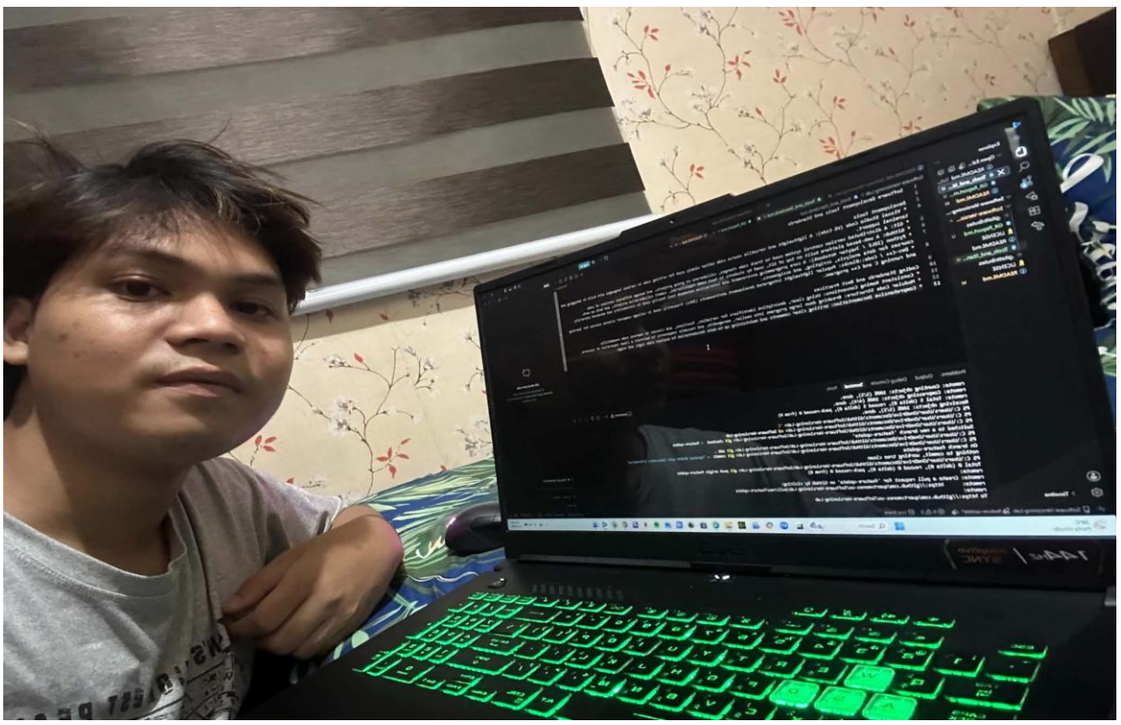
D. Findings, Observations, and Comments:

Guide Questions:

1. What additional steps are taken in this activity?
2. Are there any difficulties during the GitHub setup?
3. What are your findings, observations, and comments in this exercise?

Throughout this activity, I walked through the complete version control workflow, which involved initializing a remote GitHub repository, cloning it to my local machine, and setting up a dedicated workspace. Specifically, I created and switched to a feature-update branch, modified the project's documentation file to include my student details, staged those changes, committed them with a descriptive message, and pushed the updated branch back to the remote server. I did run into a few hurdles along the way, particularly when dealing with synchronization issues and getting GitHub's interface to recognize my branch updates for the pull request. Navigating those roadblocks required careful troubleshooting, such as verifying my branch status, managing my commits, and making sure the file modifications were correctly registered before opening the pull request to merge everything back into the main branch. Overall, this exercise gave me a much deeper, hands-on appreciation for how Git and GitHub operate in real-world scenarios. It showed me firsthand how essential version control is for tracking project history, avoiding code conflicts, and safely managing collaborative updates step by step.

PICTURE/PROOF OF DOING THE ACTIVITY



MAPÚA UNIVERSITY

Laboratory Report 1

Software Relevant Tools, Standards, and Code Versioning using Github

E. Score Sheet

Criteria	Poor (25%)	Fair (50%)	Good (75%)	Excellent (100%)	Score
I. Completeness, Documentation, and Organization of Report	The report was incomplete, messy, and had many unanswered questions.	The report was complete, messy, and had many unanswered questions.	The report was complete, neat, and had 1 or 2 unanswered questions.	The report was complete, neat, and all the questions had been answered.	
II. Coding Design and Patterns	The program had inappropriate elements and structures, incorrect indention, and poor program documentation.	The program had corrected indention, but inappropriate elements and structures and poor program documentation.	The program had corrected indention, adequate program documentation, but inappropriate elements and structures.	The program had appropriate elements and structures, correct indention, and excellent program documentation.	
III. Findings, Observations, and Comments	The findings. observations, and comments were not based on the gathered data and results. All thoughts were inconsistent and unclear.	The findings. observations, and comments based on the gathered data and results but were not elaborated. Not all thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were mostly elaborated. Most of the thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were completely elaborated. All the thoughts were consistent and clear.	
IV. Grammar and Composition	The words used were inappropriate, wrong grammar usage, and poor sentence construction was observed.	The words used were appropriate; however, bad grammar and poor sentence construction were observed.	The words used were appropriate, had proper grammar usage, but poor sentence construction was observed.	The words used were appropriate, had proper grammar usage, and excellent sentence construction.	
SCORE:					

CHECKED			
Rating		**Signature**	
		Date	